

3RD YEAR FIRST SEMESTER

CSE 3101: Computer Graphics

Credit: 3.0

Pre-requisite: MAT 1241 (Coordinate Geometry & Vector Analysis), MAT 2241 (Linear Algebra and Complex Variable)

Course Content:

Scan Conversion: Points plotting techniques, Line drawing algorithms: Horizontal and vertical line drawing algorithm, Direct Line drawing algorithm, DDA algorithm, Bresenham's algorithm. Circle Drawing algorithms: Direct method, Mid-point circle drawing algorithm. Ellipse generating algorithm. Polygon Filling: Scan line polygon fill algorithm, Boundary fill algorithm, and Flood fill algorithm.

2D and 3D transformation: Basic transformations (2D and 3D): Translation, Scaling, Rotation, Other transformations (2D and 3D): Reflection, Reflection through an arbitrary plane, Rotation about an arbitrary axis in space, shearing. Matrix representations and Homogeneous coordinates, General parallel projection Transformation.

Viewing and Clipping: Viewing pipeline, Window to viewport coordinate transformation, 3D viewing, View port clipping, clipping operations: point clipping, line clipping, polygon clipping etc.

Curves: Curve representation, Surfaces design, Hermit curves, Bezier curves, B-spline curves, End conditions for periodic B-spline curves, Rational B-spline curves.

Recommended Books:

1. Donald Hearn and Pauline Barker M. :Computer Graphics, Prentice Hall [Text Book]

2. Steven Harrington :Computer Graphics: A Programming Approach, McGraw-Hill College
3. F.S. Hill :Fundamentals of Computer Graphics, Prentice Hall

CSE 3102: Computer Graphics Lab

Credit: 1.0

Pre-requisite: MAT 1241 (Coordinate Geometry & Vector Analysis), MAT 2241 (Linear Algebra and Complex Variable)

Course Content: Using different primitives and drawing scenery using shapes Implementation of line, circle, ellipse drawing algorithms in drawing primitive shapes. Apply Different kind of geometric transformations, Coordinate transformations, Composite transformations in both 2D and 3D objects. Practice Clipping algorithms: Line, Polygon clipping Implementation of animation on 2D screen

Recommended Books:

1. Donald Hearn and Pauline Barker :Computer Graphics, Prentice Hall [Text Book]
2. Steven Harrington :Computer Graphics: A Programming Approach, McGraw-Hill College
3. F.S. Hill :Fundamentals of Computer Graphics, Prentice Hall

CSE 3103: Database Management System

Credit: 3.0

Pre-requisite: None

Course Content:

Introduction: Data and information, Purpose of Database Systems, View of Data, Database Languages DDL, DML, DCL.

Database Design and the E-R Diagram: Entity and attributes, Entity-Relationship Model, Entity-Relationship Diagrams, convert E-R model into tables.

Relational Data Model: Structure of Relational Databases, Database Schema, Keys, Schema, Diagrams, Relational Query Languages, Relational Database Design, Normalizing database up to 3rd normal form, BCNF.

SQL: Overview of the SQL Query, Language, SQL Data Definition, Basic Structure of SQL, Queries, Aggregate Functions, Modification of the Database, Join Expressions, Views, Transactions, Integrity Constraints, Functions and Procedures, Triggers, Recursive Queries, Advanced Aggregation Features.

Storage and File Structure: Overview of Physical Storage Media, Magnetic Disk and Flash Storage, RAID, Tertiary Storage, File Organization, Organization of Records in Files, Data-Dictionary Storage, Database Buffer.

Transactions: Transaction Concept, A Simple Transaction Model, Storage Structure, Transaction Atomicity and Durability, Transaction Isolation, Transaction Isolation and Atomicity, Transaction Isolation Levels, Implementation of Isolation Levels, Transactions as SQL Statements, Lock-Based Protocols, Deadlock Handling, Multiple Granularity, Timestamp-Based Protocols, Validation-Based Protocols.

Database-System Architectures: Centralized and Client-Server Architectures, Server System Architectures, Parallel Systems, Design and types of Parallel Systems, Distributed Systems architecture and its application.

Books Recommended:

1. A. Silberschatz : **Database System Concepts, McGraw-Hill.**
2. R. Ramakrishnan, : **Database Management System, McGraw-Hill**
Johannes Gehrke *Higher Education*
3. James Martin : **Principles of Database Management, Prentice-hall**
4. Ullman : **Database Management systems, Prentice-Hall Publication.**

CSE 3104: Database Management System Lab

Credit: 1.5.

Pre-requisite: None

Course Content: Create a relational database for a domain like an organization and discuss how the schema is effective for that organization. Design an E-R diagram and mapping a relational model from that E-R diagram for a given scenario. Practice intermediate and advanced SQL operation for a given relation. o Intermediate SQL: Join Expressions, Views etc. o Advanced SQL: Trigger, Stored procedure, sub query etc. Practice cascading operation using keys for the given relations. Removing insertion, deletion and update anomalies using normalization technique and redesign the database schema.

Books Recommended:

1. A. Silberschatz : **Database System Concepts, McGraw-Hill.**
2. James Martin : **Principles of Database Management, Prentice-hall**
3. Ullman : **Database Management systems, Prentice-Hall Publication.**
4. Abey : **Oracle 8i a Beginners Guide, McGraw-Hill**

CSE 3105: Computer Architecture

Credit: 3.0

Pre-requisite: CSE 2105 (Digital System Design)

Course Content:

Basic Concept: Introduction to Computer Architecture, Computer Design and Computer Organization. Interaction of hardware with the system and application software, basic hardware components of computer system and bus structure, concept and analysis of performance.

Machine Instruction: Concept of machine instruction, different parts of machine instruction, memory model maps for instruction, micro-operation, registers, addressing modes, I/O registers and instruction, program control status bits.

Processing Unit: Components of processing units and their functions, general register organization for micro-operation, control word, pipelining, instruction pipelining and performance of pipelining.

I/O Organization: Peripheral devices, I/O Interface, I/O bus module, synchronous data transfer and its different types, asynchronous data transfer, different modes of transfer, interrupt processing, direct memory access (DMA), I/O processor (IOP).

Memory Organization: Concept of memory, memory hierarchy, main memory, functions of RAM and ROM, memory address map, memory connection with CPU, Cache memory and its performance, secondary storage, virtual memory.

Composite Architecture: Multiprocessor and its different types, multicomputer, CISC and RISC processors, parallel processing, parallelism in multiprocessor, Flynn's classification.

Books Recommended:

1. Carl Hamacher, Zvonko Vranesic : **Computer Organization,**

2. Patterson, David A. and John L. Hennessy

Hill. [Text Book]

:Computer Organization and Design: The Hardware/Software Interface, Third edition.

3. William Stallings

:Computer Organization and Architecture: Designing for Performance, Prentice Hall

CSE 3106: Computer Architecture Lab

Credit: 1.0

Pre-requisite: None

Course Content: Design asynchronous counters using J-K flip-flop, Design synchronous counters to count the sequence using J-K flipflop Design 8-bit Serial In/ Parallel Out shift register using 74LS164, Design 8 bit Parallel In/ Serial Out shift register using 74LS165 Design memory read/write operation using IDT 6116 Static RAM.

Books Recommended:

1. Carl Hamacher, Zvonko Vranesic and Safwat Zaky

:Computer Organization, McGraw-Hill. [Text Book]

2. Patterson, David A. and John L. Hennessy

:Computer Organization and Design: The Hardware/Software Interface, Third edition.

3. William Stallings

:Computer Organization and Architecture: Designing for Performance, Prentice Hall

CSE 3107: Communication Engineering

Credit: 3.0

Pre-requisite: CSE 2207 (Computer Networks)

Course Content:

Communication Engineering Fundamentals: Waveforms Spectra, Elements of basic communication systems, Periodic and nonperiodic waveforms and their properties, Fourier series, Noise and its different types.

Analogue Communication: Amplitude modulation, Amplitude modulation index, Frequency spectrum for sinusoidal AM, Frequency Modulation: Frequency Modulation, Sinusoidal FM, Frequency spectrum for Sinusoidal FM, Phase Modulation. Pulse Modulation, Pulse Codes Modulation (PCM), Quantization, Compression, PCM Receiver, Differential PCM, Delta Modulation, Sigma-Delta A/D conversion, Pulse Modulation. Analog Multiplexing

Digital Communication: Basic Digital Communication System, Synchronization, Asynchronous Transmission, Probability of Bit Error in Base band Transmission, Matched Filter, Eye Diagrams, Digital Carrier Systems, Amplitude Shift keying, Frequency Shift Keying, Phase Shift Keying, Carrier Recovery Circuits, Quadrature Phase Shift Keying, Error Control Coding, Block Control, Repetition Encoding, Parity Encoding, Convolution Encoding. Digital Multiplexing.

Satellite Communication: Satellite Communication, Kepler's First and Second Law, Orbits, Geostationary Orbits, Power System. 5G technologies.

Fiber Optic Communication: Fiber Optic Communication, Propagation within a Fiber, Modes of Propagation, Losses in Fibers, Light sources for Fiber optics, Photo detectors.

Books Recommended:

1. Behrouz A. Forouzan : **Data Communication and Networking, Tata McGraw-Hill Edition [Text Book]**
2. George Kennedy, Bernard Davis : **Electronic communication System, Tata McGraw-Hill Edition [Text Book]**
3. John M. Senior : **Optical Fiber Communication, Prentice-Hall of India Pvt Ltd**
4. William Stallings : **Data and computer Communication, Addison Wesley**
5. Theodore S. Rappaport : **Wireless Communication, Prentice-Hall of India Pvt Ltd**

MAT 3141: Applied Statistics and Probability

Credit: 3.0

Pre-requisite: MAT 1141 (Differential and Integral Calculus), MAT 2241

(Linear Algebra and Complex Variables)

Course Content:

An introduction: Definition of Statistics, Uses and importance, Variables, Measurement level of variables, Sample and Population.

Measures of central tendency: Introduction, the arithmetic mean, Geometric mean, Harmonic mean, Median, mode, computing of means for grouped data and ungrouped data, Advantages, and disadvantages, and Uses of all measures.

Measures of dispersion: Introduction of dispersion, Absolute and Relative measures, Range, mean deviation, Standard deviation, Variance, Quartile deviation, Coefficient of Variation, Advantages, disadvantages and uses of all measures.

Data representation: Classification, Types of classification, Tabular representation, Graphical representation, Pie chart, Histogram, Bar chart, Frequency curve, Frequency Polygon and Ogive curve.

Correlation and Regression: Correlation, Types of correlation, Importance and uses of correlation, Properties of correlation, Rank correlation, Regression, Differences between regression line and regression model, Simple linear Regression, Importance of regression analysis, Comparison of regression and correlation.

Probability: Concept of probability, Sample Space, Events, Union and Intersection of Events, Probability of events, Laws of probability, Conditional probabilities, Bayes' theorem, Chebyshev's Inequality and Practical examples.

Random variables and probability Distribution: Basic concepts, Discrete and continuous random variables, Probability Mass Function (PMF), Probability Density Function (PDF), Cumulative Distribution Function (CDF), Mathematical expectation and variance, Joint, marginal and conditional density functions, Conditional Expectation and conditional variance, Probability generating function, Characteristic function, Binomial distribution, Poisson distribution, and Normal distribution with Practical examples.

Sampling design and Statistical test: Sampling Design, Simple random sampling, Systematic sampling, Stratified sampling, Cluster sampling, Judgment sampling, Chi-squared test, Z test, t-test and F-test.

Books Recommended:

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| 1. Murray R. Spiegel and Larry J. Stephens | :Schaum's Outlines of Statistics, Companies, USA, Fourth Editions. [Text Book] |
| 2. Mr. Nurul Islam | :Statistics & Probability |
| 3. J. B. Anderson | :Interpreting Data, Chapman and Hall, London |
| 4. H. Cramer | :The Elements of Probability Theory, Wiley, N. Y |
| 5. P. Hoel | :Introductory Statistics, Wiley and Sons, N. Y |
| 6. Lind, Marchal, Mason | :Statistical Techniques in Business & Economics, Irwin/McGraw-Hill, USA, Fourteen Editions. |